

From an intrinsic to a fictitious vector potential: exact electron factorization as a route beyond TDDFT

Foundations and Extensions of DFT 2026

Sara Giarrusso

Technical University of Eindhoven



April 21, 2026

Outline

1. Exact electron factorization (EEF) over the years
2. An effective equation for $\sqrt{n(\mathbf{r}, t)}$
 - ▶ Intrinsic vector potential $\mathbf{A}(\mathbf{r}, t)$ and current density
3. Matching both n and $\mathbf{j}$ via non-interacting auxiliary system
4. Conclusive thoughts

Exact (electron) factorization $\Psi = \chi\Phi$



Interacting system

$$\hat{H}^N \Psi = i \partial_t \Psi$$

Hamiltonian partition

$$\hat{H}^N = \hat{h}_1(\mathbf{r}_1, \dots, \mathbf{r}_N, t) + \hat{H}^{N-1}(\mathbf{r}_2, \dots, \mathbf{r}_N, t)$$

where

$$\hat{h}_1 = -\frac{\nabla_{\mathbf{r}}^2}{2} + \sum_{i=2}^N \frac{1}{r_{1i}} + v(\mathbf{r}, t)$$

Focus on one reference electron

$$\mathbf{r} = \mathbf{r}_1$$

Exact factorization of the TD wavefunction

$$\Psi(\mathbf{x}_1, \dots, \mathbf{x}_N, t) = \chi(\mathbf{r}, t) \Phi(\sigma, \mathbf{x}_2 \dots \mathbf{x}_N; \mathbf{r}, t)$$

Partial normalization condition (PNC)

$$\sum_{\sigma} \int |\Phi|^2 d\mathbf{x}_2 \dots d\mathbf{x}_N = 1 \quad \forall \mathbf{r}, t$$

Determines χ and Φ up to a phase factor

$$|\chi|^2 = \frac{n}{N} \quad |\Phi|^2 = \frac{N|\Psi|^2}{n}$$

Gauge choice

$$\chi(\mathbf{r}, t) = \sqrt{\frac{n(\mathbf{r}, t)}{N}}$$

Emergent potentials

Projecting the TDSE onto χ yields an equation for the marginal amplitude:

$$\langle \Phi | \frac{\partial}{\partial t} (\chi \Phi) \rangle_{\setminus \{\mathbf{r}, t\}} = \langle \Phi | \left(\hat{h}_1 + \hat{H}^{N-1} \right) (\chi \Phi) \rangle_{\setminus \{\mathbf{r}, t\}}$$

The PNC implies that Dirac brackets of space/time derivatives wrt reference electron are purely imaginary, defining real-valued emergent potentials:

Vector potential

$$\mathbf{A}(\mathbf{r}, t) = -i \langle \Phi | \nabla_{\mathbf{r}} \Phi \rangle_{\setminus \{\mathbf{r}, t\}}$$

Scalar potential

$$v_{\text{roc}}(\mathbf{r}, t) = -i \langle \Phi | \partial_t \Phi \rangle_{\setminus \{\mathbf{r}, t\}}$$

Orbital-free TDDFT equation

Marginal equation

$$(v_{\text{cond}} + v_{\text{kin}} + \tilde{v}_{N-1} + v_{\text{roc}} + v) \chi - \frac{\nabla^2}{2} \chi - \left(i\mathbf{A} \cdot \nabla + \frac{i}{2} \nabla \cdot \mathbf{A} \right) \chi = i\partial_t \chi$$

- Represents an **orbital-free TDDFT formulation**.

If the potentials were known density functionals, a direct propagation of $n(\mathbf{r}, t)$ would be possible.

Interacting scalar potentials

The additional scalar potentials appearing in the marginal equation are defined as:

$$v_{\text{cond}}(\mathbf{r}, t) = \sum_{i=2}^N \left\langle \Phi \left| \frac{1}{r_{1i}} \right| \Phi \right\rangle_{\setminus \{\mathbf{r}, t\}}$$

$$v_{\text{kin}}(\mathbf{r}, t) = \frac{1}{2} \langle \nabla_{\mathbf{r}} \Phi | \nabla_{\mathbf{r}} \Phi \rangle_{\setminus \{\mathbf{r}, t\}}$$

$$\tilde{v}_{N-1}(\mathbf{r}, t) = \langle \Phi | \hat{H}^{N-1} | \Phi \rangle_{\setminus \{\mathbf{r}, t\}}$$

These are the time-dependent counterparts of the static EEF potentials.

Current-carrying eigenstate

$$\hat{H}^N \Psi_m = E_m^N \Psi_m$$

$$\Psi_m(1, \dots, N) = \underbrace{\chi_m(\mathbf{r})}_{\downarrow \sqrt{n_m(\mathbf{r})/N}} \Phi_m(\sigma, 2, \dots, N; \mathbf{r})$$

$$\left[\left(v_{\text{cond},m} + v_{\text{kin},m} + v_{N-1,m} + v - \frac{\nabla_{\mathbf{r}}^2}{2} \right) - i \left(\mathbf{A}_m \cdot \nabla_{\mathbf{r}} + \frac{1}{2} \nabla_{\mathbf{r}} \cdot \mathbf{A}_m \right) \right] \chi_m = -I_{p,m} \chi_m$$

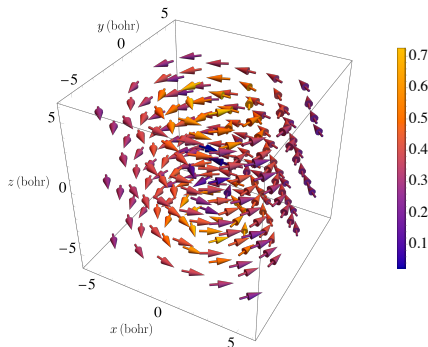
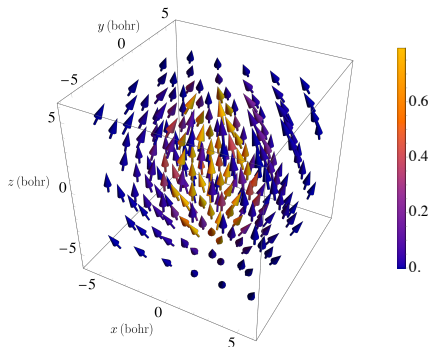
with

$$\mathbf{A}_m(\mathbf{r}) = -i \langle \Phi_m | \nabla_{\mathbf{r}} \Phi_m \rangle_{\setminus \{\mathbf{r}\}}$$

S.G., P. Gori-Giorgi, F. Agostini, ChemPhysChem, e202400127 (2024)

$N = 2$ non-interacting fermions in the Coulomb potential

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}} (\psi_{100}(\mathbf{r}_1)\psi_{211}(\mathbf{r}_2) - \psi_{211}(\mathbf{r}_1)\psi_{100}(\mathbf{r}_2))$$

 $\mathbf{A}(x, y, z)$  $\nabla \times \mathbf{A}(x, y, z)$ 

Current and continuity

With the chosen gauge, the intrinsic vector potential relates to the paramagnetic current density as:

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mathbf{j}_{\Psi,p}}{n}(\mathbf{r}, t)$$

with

$$\mathbf{j}_{\Psi,p} = N \operatorname{Re} \langle \Psi | -i \nabla_{\mathbf{r}} | \Psi \rangle_{\setminus \{\mathbf{r}, t\}}$$

Since no *external* vector potential is present, $\mathbf{j}_{\Psi,p} = \mathbf{j}$, and the **imaginary part** reduces to the **continuity equation**

$$\partial_t n = -\nabla \cdot \mathbf{j}$$

Real and imaginary parts

Separating real and imaginary parts of the marginal equation:

Real part

$$(v_{\text{cond}} + v_{\text{kin}} + \tilde{v}_{N-1} + v_{\text{roc}} + v)(\mathbf{r}, t) = \frac{\nabla^2 \chi}{2\chi}(\mathbf{r}, t)$$

Imaginary part

$$-\left(\mathbf{A} \cdot \nabla_{\mathbf{r}} + \frac{\nabla_{\mathbf{r}} \cdot \mathbf{A}}{2}\right)\chi = \frac{\partial}{\partial t}\chi$$

Magnetic-like vector potential

$\mathbf{A}$ enters the marginal equation as an **external vector potential** in (TD)CDFT

$$\frac{[-i\nabla_{\mathbf{r}} + \mathbf{A}(\mathbf{r}, t)]^2}{2}\chi + \tilde{v}\chi = i\partial_t\chi \qquad \frac{[-i\nabla_{\mathbf{r}} + \mathbf{A}_m(\mathbf{r})]^2}{2}\chi_m + \tilde{v}_m\chi_m = -I_{p,m}\chi_m$$

As a consequence, the current splits as:

Full wavefunction Ψ

$$\begin{cases} \mathbf{j}_{\Psi, \text{para}} = n\mathbf{A} \\ \mathbf{j}_{\Psi, \text{dia}} = 0 \end{cases}$$

Marginal amplitude χ

$$\begin{cases} \mathbf{j}_{\chi, \text{para}} = 0 \\ \mathbf{j}_{\chi, \text{dia}} = n\mathbf{A} \end{cases}$$

Same total current, different para/dia decomposition.

Motivation

- presence of an intrinsic vector potential in the equation for $\sqrt{n}$
- non v -representability of current densities (continuity equation determines only longitudinal current)
- memory-dependent functionals using an XC vector potential more “natural” /easier (case of memory-dependent LDA)

G. Vignale, W. Kohn, PRL, **77**.10: 2037, (1996)

- ...

Hypothesis

- ▶ could a TD Kohn-Sham system with an external **vector** potential $\mathbf{A}_s$ be more successful than standard TDDFT for non-local effects?

Non-interacting auxiliary system

$$\{v_s, \mathbf{A}_s\} \implies \{n, \mathbf{j}\}$$

Auxiliary Hamiltonian

$$\hat{H}_s^N = \sum_i \left[\frac{(-i\nabla + \mathbf{A}_s)^2}{2} + v_s \right]$$

Wavefunction factorization (same marginal amplitude by construction)

$$\Psi_s = \chi(\mathbf{r}, t) \Phi_s(\sigma, \mathbf{x}_2, \dots, \mathbf{x}_N; \mathbf{r}, t)$$

Hamiltonian partition

$$\hat{H}_s^N = \hat{h}_s(\mathbf{r}_1, t) + \hat{H}_{s*}^{N-1}(\mathbf{r}_2, \dots, \mathbf{r}_N, t)$$

v_s and $\mathbf{A}_s$ in $\hat{H}_{s*}^{N-1}$ come from the N -electron system.

Emergent potentials of the auxiliary system

By analogy with interacting case, define following quantities from the non-interacting conditional amplitude Φ_s :

$$\mathbf{A}_{\Phi_s}(\mathbf{r}, t) = -i \langle \Phi_s | \nabla_{\mathbf{r}} | \Phi_s \rangle_{\setminus \{\mathbf{r}, t\}}$$

$$v_{s, \text{roc}}(\mathbf{r}, t) = -i \langle \Phi_s | \partial_t | \Phi_s \rangle_{\setminus \{\mathbf{r}, t\}}$$

$$v_{s, \text{kin}}(\mathbf{r}, t) = \frac{1}{2} \langle \nabla_{\mathbf{r}} \Phi_s | \nabla_{\mathbf{r}} | \Phi_s \rangle_{\setminus \{\mathbf{r}, t\}}$$

$$\tilde{v}_{s, N-1}(\mathbf{r}, t) = \left\langle \Phi_s \left| \hat{H}_{s*}^{N-1} \right| \Phi_s \right\rangle_{\setminus \{\mathbf{r}, t\}}$$

Real and imaginary parts (auxiliary system)

Real part

$$(v_{s,\text{kin}} + \tilde{v}_{s,N-1} + v_{s,\text{roc}} + \tilde{v}_s + \mathbf{A}_s \cdot \mathbf{A}_{\Phi_s}) \chi = \frac{\nabla^2 \chi}{2\chi}$$

Imaginary part

$$-\left(\mathbf{A}_{\Phi_s} \cdot \nabla_{\mathbf{r}} + \frac{\nabla_{\mathbf{r}} \cdot \mathbf{A}_{\Phi_s}}{2} + \frac{\nabla_{\mathbf{r}} \cdot \mathbf{A}_s}{2} + \mathbf{A}_s \cdot \nabla_{\mathbf{r}}\right) \chi = \frac{\partial}{\partial t} \chi$$

where $\tilde{v}_s = v_s + |\mathbf{A}_s|^2/2$ absorbs the scalar term from $\mathbf{A}_s$.

Current in the auxiliary system

By definition of $\mathbf{A}_{\Phi_s}$:

$$\mathbf{A}_{\Phi_s}(\mathbf{r}, t) = \frac{\mathbf{j}_{\psi_s, p}(\mathbf{r}, t)}{n}$$

Because the auxiliary system has an external vector potential $\mathbf{A}_s$, the **total current** includes a **diamagnetic contribution**:

$$\mathbf{j}(\mathbf{r}, t) = (\mathbf{A}_{\Phi_s}(\mathbf{r}, t) + \mathbf{A}_s(\mathbf{r}, t)) n(\mathbf{r}, t)$$

The imaginary part of the auxiliary marginal equation then reduces again to the continuity equation, consistently.

Auxiliary scalar potential v_s

Equating real parts of the interacting and auxiliary marginal equations gives :

$$v_s(\mathbf{r}, t) = \left(v_{\text{cond}} + [v_{\text{kin}} - v_{s,\text{kin}}] + [\tilde{v}_{N-1} - \tilde{v}_{s,N-1}] \right. \\ \left. + [v_{\text{roc}} - v_{s,\text{roc}}] - \frac{|\mathbf{A}|^2 - |\mathbf{A}_{\Phi_s}|^2}{2} + v \right)(\mathbf{r}, t)$$

To be contrasted with the static decomposition

$$v_s(\mathbf{r}) = (v_{\text{cond}} + [v_{\text{kin}} - v_{s,\text{kin}}] + [v_{N-1} - v_{s,N-1}] + v)(\mathbf{r})$$

M. A. Buijse, E. J. Baerends, and J. G. Snijders, PRA, **40**, 4190-4202 (1989)

and known relations

$$E_{\text{Hxc}} = \int \left(\frac{v_{\text{cond}}}{2} + [v_{\text{kin}} - v_{s,\text{kin}}] \right) n$$

$$\frac{\delta E_{\text{Hxc}}}{\delta n} = v_{\text{cond}} + [v_{\text{kin}} - v_{s,\text{kin}}] + [\tilde{v}_{N-1} - \tilde{v}_{s,N-1}]$$

Auxiliary vector potential $\mathbf{A}_s$

Comparing the non-interacting with the interacting result for $\mathbf{j}$ gives:

$$\mathbf{A}_s(\mathbf{r}, t) = (\mathbf{A} - \mathbf{A}_{\Phi_s})(\mathbf{r}, t)$$

- ▶ structure of a **correlation-only vector potential** (difference between interacting and non-interacting quantities)
- ▶ no ground-state counterpart! no prior knowledge

Outlook: numerical inversion

Determine $\{v_s, \mathbf{A}_s\}$ for at least one simple TD system

- Determine quantities of interest for a TD interacting system
- Extract

$$n(\mathbf{r}, t), \quad \mathbf{j}(\mathbf{r}, t)$$

and

$$\{v_{\text{cond}}, v_{\text{kin}}, \tilde{v}_{N-1}, v_{\text{roc}}, \mathbf{A}(\mathbf{r}, t)\}$$

from accurate wavefunctions

- From initial guess of ψ_j , construct $\{v_s, \mathbf{A}_s\}$ self-consistently via the equations:

$$\left(\frac{(-i\nabla_{\mathbf{r}} + \mathbf{A}_s)^2}{2} + v_s \right) \psi_j(\mathbf{r}, t) = i\partial_t \psi_j(\mathbf{r}, t)$$

Similar in spirit to the static inversion scheme in [Cuevas-Saavedra, P. W. Ayers, V. N. Starovero, JCP, **143**, 244116 \(2015\)](#) but now time propagation

Summary

- proposed TD “current” DFT for electronic dynamics in absence of external magnetic field
- time-independent and time-dependent problems on equal footing
- decomposition of v_s and $\mathbf{A}_s$ in terms of interacting and non-interacting states (one- and two-body density matrices)

Some open questions

- What is the relation between $\{v_s, \mathbf{A}_s\}$ and the action functional?
- What role does $\mathbf{A}_s$ play in the time evolution?
- What happens to the boundary term of TDDFT, $\langle \Psi | \delta \Psi \rangle$? Can it be expressed in terms of local scalar and vector potentials?
- What exact conditions does this framework entail (connection to force-balance approach)?

Summary

- proposed TD “current” DFT for electronic dynamics in absence of external magnetic field
- time-independent and time-dependent problems on equal footing
- decomposition of v_s and $\mathbf{A}_s$ in terms of interacting and non-interacting states (one- and two-body density matrices)

Some open questions

- What is the relation between $\{v_s, \mathbf{A}_s\}$ and the action functional?
- What role does $\mathbf{A}_s$ play in the time evolution?
- What happens to the boundary term of TDDFT, $\langle \Psi | \delta \Psi \rangle$? Can it be expressed in terms of local scalar and vector potentials?
- What exact conditions does this framework entail (connection to force-balance approach)?

Thank you!